

Document	OSIM-RPT-005	Status	READY FOR SIMULATION	Version	1.0
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1. Lab Build Summary

The OSIM Lab was designed and built from scratch in approximately 4 days. The environment implements the OSIM Framework (Orthogonal Systems Invariance Method) v1.0 — a Zero Trust security architecture methodology with machine-verifiable invariants. The lab is now fully operational and ready for adversary simulation exercises. This document captures the complete pre-simulation state as a baseline before the first ATT&CK; operation is executed.

2. Full Infrastructure Inventory

2.1 Hardware

Device	CPU	RAM	Storage	Role	Status
T620	—	8GB	64GB	OPNsense 26.1.5 firewall	Production ✓
P520	Xeon W-2135	64GB	NVMe	Proxmox blue node	Production ✓
T480	i7-8650U	48GB	SSD	Proxmox red node	Pending deploy
Cisco 3560CX	—	—	—	Primary managed switch	Production ✓
Cisco SF300	—	—	—	Secondary switch	Production ✓
Debian box	Ryzen 5	16GB	256GB	Wiki.js knowledge base	Production ✓
MacBook M1	M1	32GB	500GB	Management workstation	Production ✓

2.2 Network Architecture

VL AN	Name	Subnet	Purpose	Status
10	MGMT-A	10.10.10.0/24	Proxmox/infra admin	Active ✓
11	MGMT-B	10.10.11.0/24	MacBook management	Active ✓
20	LAB-OPS	10.10.20.0/24	TheHive, MISP	Active ✓
30	WINDOWS-ENT	10.10.30.0/24	AD DC, workstations	Active ✓

40	SECURITY	10.10.40.0/24	Wazuh, Greenbone, Velociraptor, Caldera	Active ✓
50	RED-OPS	10.10.50.0/24	T480 attacker tooling	Pending
60	C2-ISOLATED	10.10.60.0/24	Sliver/Havoc C2 listeners	Pending
70	MAL-SANDBOX	10.10.70.0/24	Malware detonation	Pending
80	FAKE-INTERNET	10.10.80.0/24	INetSim fake internet	Pending
90	OSWE-WEB	10.10.90.0/24	Web app targets	Pending
91	OSED-TARGET	10.10.91.0/24	Exploit dev targets	Pending
92	JUMP-ADMIN	10.10.92.0/24	Hardened jump host	Pending

2.3 VM Inventory — P520 Blue Node

VMI D	Name	OS	RAM	IP	VLAN	Role	Status
100	wazuh-manager	Ubuntu 22.04	16GB	10.10.40.10	40	Wazuh SIEM	Active ✓
101	thehive	Ubuntu 22.04	4GB	10.10.20.10	20	Case management	Active ✓
102	misp	Ubuntu 22.04	4GB	10.10.20.11	20	Threat intelligence	Active ✓
103	greenbone	Ubuntu 22.04	4GB	10.10.40.20	40	Vuln scanner	Active ✓
104	velociraptor	Ubuntu 22.04	4GB	10.10.40.30	40	Live response	Active ✓
105	caldera	Ubuntu 22.04	4GB	10.10.40.40	40	Adversary simulation	Active ✓
106	win-dc01	WS 2022	6GB	10.10.30.10	30	AD DC — osim.lab	Active ✓
107	win-client01	Windows 11	4GB	10.10.30.20	30	Domain workstation	Active ✓
108	win-client02	Windows 10	4GB	10.10.30.21	30	Domain workstation	Active ✓

3. Detection Stack — Pre-Simulation Baseline

Tool	Version	URL	Connected Agents/Clients	Status
Wazuh SIEM	4.11.2	https://10.10.40.10	4: OPNsense, DC01, CLIENT01, CLIENT02	Active ✓
TheHive	5.4	http://10.10.20.10:9000	MISP integration active	Active ✓
MISP	2.4	https://10.10.20.11	TheHive integration active	Active ✓
Greenbone	26.21.0	https://10.10.40.20:9392	0 scans run — clean baseline	Active ✓
Velociraptor	0.76.1	https://10.10.40.30:8889	3: DC01, CLIENT01, CLIENT02	Active ✓
Caldera	Latest	http://10.10.40.40:8888	3 Sandcat agents: DC01, CLIENT01, CLIENT02	Active ✓
Sysmon	v15.90	All Windows hosts	Olaf Hartong modular config	Active ✓

4. Active Directory — Pre-Simulation Baseline

Component	Value
Domain	osim.lab (NetBIOS: OSIM)
Domain Controller	DC01.osim.lab — 10.10.30.10 — WS2022
Workstations	CLIENT01 (Win11, 10.10.30.20), CLIENT02 (Win10, 10.10.30.21)
Users	jsmith, sjones, mchen, lbrown, tdavis (OSIM-Users) helpdesk (Domain Admin) svc-backup, svc-sql (ServiceAccounts)
Sysmon	v15.90 Olaf Hartong modular config on DC01, CLIENT01, CLIENT02
Wazuh agents	Active on DC01, CLIENT01, CLIENT02 → reporting to 10.10.40.10
Velociraptor	Agents active on DC01, CLIENT01, CLIENT02 → 10.10.40.30:8000
Caldera Sandcat	Agents active on DC01, CLIENT01, CLIENT02 → 10.10.40.40:8888
Windows Defender	Disabled on all Windows hosts for lab simulation purposes

5. First ATT&CK; Simulation Plan

The first adversary simulation will be executed using Caldera with Sandcat agents on the Windows domain hosts. The objective is to validate the full detection pipeline: Caldera executes techniques → Sysmon captures telemetry → Wazuh generates alerts → TheHive receives case. This validates OSIM invariant: detection without containment is forensics, not security.

Step	ATT&CK; Technique	Tool	Target	Expected Detection
1	T1059.001 — PowerShell	Caldera	DC01	Wazuh rule 92000+ — PowerShell execution
2	T1057 — Process Discovery	Caldera	DC01	Sysmon Event ID 1 — process creation
3	T1082 — System Info Discovery	Caldera	CLIENT01	Sysmon Event ID 1 — systeminfo.exe
4	T1083 — File Discovery	Caldera	CLIENT02	Sysmon Event ID 1 — dir command
5	T1016 — Network Discovery	Caldera	DC01	Sysmon Event ID 3 — network connection

6. Snapshot Registry — Pre-Simulation

VMI D	VM	Snapshot	State
100	wazuh-manager	—	Running — no snapshot needed, stateful SIEM
103	greenbone	greenbone-docker-running	16 containers running, no scans yet
104	velociraptor	velociraptor-agents-connected	3 Windows agents connected
105	caldera	caldera-running	Systemd service, 3 Sandcat agents connected
106	win-dc01	dc01-caldera-sandcat	Sysmon+Wazuh+Velociraptor+Sandcat all active
107	win-client01	win11-caldera-sandcat	Sysmon+Wazuh+Velociraptor+Sandcat all active
108	win-client02	win10-caldera-sandcat	Sysmon+Wazuh+Velociraptor+Sandcat all active

7. Pending Items — Post-Simulation

Priority	Item	Description
HIGH	T480 red node	Connect to switch, configure Proxmox, deploy Kali Purple VLAN 50
HIGH	First Greenbone scan	Scan 10.10.30.0/24, establish vuln baseline before red team
HIGH	OSIM v2.0 declaration	After first successful ATT&CK; sim with verified pipeline end-to-end
MEDIUM	CLIENT01 Sandcat persist	Make Sandcat agent persistent across reboots on Win11
MEDIUM	win-server01	Deploy Windows Server 2022 member server with IIS/MSSQL/SMB
MEDIUM	MacBook file cleanup	Organize lab files into ~/Lab/ folder structure
LOW	OpenCTI	Deploy threat intel platform for GCTI cert prep

LOW	SOAR automation	n8n or Shuffle — automate Wazuh→TheHive→MISP workflow
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